

PhantomCrypt: Second-Order Deniable Encryption with Post-Quantum Security

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Abstract. Traditional deniable encryption focuses on denying the *content* of secret communications by allowing plausible alternative plaintexts under coercion. However, the recognizable use of deniable encryption may itself defeat its purpose: any revealed plaintext becomes suspicious if the coercer detects that a non-standard encryption tool was used. We introduce *second-order deniability*: the property that the use of a deniability mechanism is itself deniable. We formalize this notion via game-based security definitions for *content deniability* (CD) and *existence deniability* (ED), prove that CD and ED are logically independent, and show that their conjunction suffices to achieve second-order deniability. We present PhantomCrypt, a concrete construction composing False-Bottom Encryption (for CD) with Invisible Encryption (for ED) under a hybrid encryption envelope, and prove that this composition preserves both security properties with an explicit security bound. A proof-of-concept implementation demonstrates practical feasibility, achieving encryption of a 32-byte message with three decoys in under 10ms on commodity hardware.

Keywords: Deniable encryption · Plausible deniability · Post-quantum cryptography · Secret sharing · Steganography

1 Introduction

The growing sophistication of state-sponsored surveillance and coercive interrogation poses significant challenges to privacy. Traditional cryptographic solutions provide confidentiality, but they fail under coercion: if an adversary forces disclosure of decryption keys, the protected content is exposed. Classical deniable encryption, introduced by Canetti et al. [11], addresses this by enabling parties under duress to present a plausible but false message. We call this property *content deniability* (CD).

However, a critical limitation remains. Specialized deniable encryption schemes produce ciphertexts that do not conform to any standard protocol format. An observer who detects data outside a recognized format can conclude that non-standard encryption is being used. In authoritarian contexts, the mere detection of unusual encryption may be treated as evidence of wrongdoing.

This gap motivates *existence deniability* (ED): the property that covert communication is indistinguishable from innocuous data, including legitimate encrypted traffic. Where CD asks “which message was encrypted?”, ED asks “was anything encrypted at all?”

We define *second-order deniability* as the property that the use of a deniability mechanism is itself deniable. First-order deniability protects the *content* of a message (CD); second-order deniability additionally protects the *existence* of the deniable communication itself (ED). A second-order deniable scheme denies both what was encrypted and the fact that deniable encryption was used.

Contributions. We formalize CD, ED, and second-order deniability via game-based definitions agnostic to any specific construction, and prove CD and ED are logically independent with their conjunction necessary and sufficient for second-order deniability. We present PhantomCrypt, composing FBE (for CD) with IE (for ED) within a hybrid PQC-KEM envelope, proving ED via reduction to KEM and AEAD security, and CD via an information-theoretic argument in the random oracle model. A two-stage composition lemma shows these properties do not degrade under sequential attack. We identify key-reuse vulnerabilities with mitigations, and demonstrate practical feasibility via a proof-of-concept implementation.

1.1 Threat Model: A Two-Stage Adversary

A natural objection is that CD and ED are redundant: if ED succeeds, no coercer suspects anything, making CD unnecessary; if ED fails, CD is the only defense. We address this with a two-stage threat model. In Stage 1 (network observation), the adversary passively observes traffic; ED protects here. If ED holds, the adversary has no reason to escalate. In Stage 2 (device seizure or coercion), the adversary escalates due to non-cryptographic intelligence (e.g., informants), seizes the device, and discovers PhantomCrypt is in use. At this point, the adversary knows a deniability mechanism was employed, so second-order deniability in its full sense no longer holds. However, CD remains intact: the adversary forces decryption, the user reveals a decoy key, and the adversary cannot identify the true message. The conjunction $CD \wedge ED$ provides defense-in-depth: ED reduces the probability of reaching Stage 2, and CD provides protection if Stage 2 occurs.

Scope. We do not claim that PhantomCrypt hides itself from software forensics on a seized device; that would require integration into a standard protocol (e.g., a TLS library), which is orthogonal to our contribution. Our ED claim is about network-level indistinguishability.

2 Conceptual Overview

PhantomCrypt achieves second-order deniability by building on a single cryptographic primitive: information-theoretically secure threshold secret sharing via Lagrange interpolation over $\mathbb{F}_p$. This primitive underpins both Invisible Encryption (IE) for existence deniability and False-Bottom Encryption (FBE) for content deniability. We explain the core idea through a concrete scenario before formalizing it.

The scenario. Sarah wishes to send a secret message m_1 to Robert. She composes a natural-language cover text $T = (w_1, \dots, w_L)$ for instance, an ordinary email or article. Using a pre-shared seed x_0 , Sarah produces a pseudorandom sequence $x_1, \dots, x_{k-1}$ by cryptographic means, and, with another pseudorandom sequence, selects $k - 1$ words $w_{i_1}, \dots, w_{i_{k-1}}$ from the text. Each selected word defines a share $s_j \leftarrow H(w_{i_j}) \in \mathbb{F}_p$, where H is a hash function modeled as a random oracle. These shares, together with the message m_1 placed at the secret point $P(0)$, determine a unique polynomial P of degree at most $k - 1$ via Lagrange interpolation through the k points,

$$(0, m_1), (x_1, s_1), \dots, (x_{k-1}, s_{k-1}). \quad (1)$$

Sarah evaluates P at a fresh abscissa x_{new} (likewise pseudorandomly derived from x_0) to obtain the transmitted share $s_{\text{new}} = P(x_{\text{new}})$. By the security of Shamir secret sharing, s_{new} on its own reveals nothing about m_1 : it is a uniformly random element of $\mathbb{F}_p$. Because it looks random, s_{new} can serve unsuspiciously as a session key in a Key Encapsulation Mechanism (KEM) exchange embedded within the transcript. The cover text is transmitted verbatim: no word is modified, and the shares are *derived from* it rather than embedded into it. This is the Invisible Encryption mechanism [2], and it provides existence deniability.

Adding content deniability. Suppose Sarah is under coercion and must explain her transcript. With IE alone, she can claim that the only secrets are the KEM session key. Without the seed x_0 or knowledge of which words serve as shares, the coercer cannot recover m_1 . However, if the coercer becomes suspicious and Sarah’s device is compromised, there is only one message to find.

Sarah has already foreseen this possibility. Before coercion, she has already embedded a second message m_2 using a different set of pseudorandomly selected words (shown with gray highlighting in Figure 1), each producing its own share. Figure 1 illustrates a concrete transcript showing both mechanisms: the final share for m_2 is carried inside the KEM exchange as the session key, and the final share for m_1 is carried in a QR code disguised as a login token.

A recent scheme called False-Bottom Encryption (FBE) [1] formalizes this multi-message capability: it embeds an arbitrary number of messages within a single ciphertext via underdetermined linear systems over $\mathbb{F}_p$.

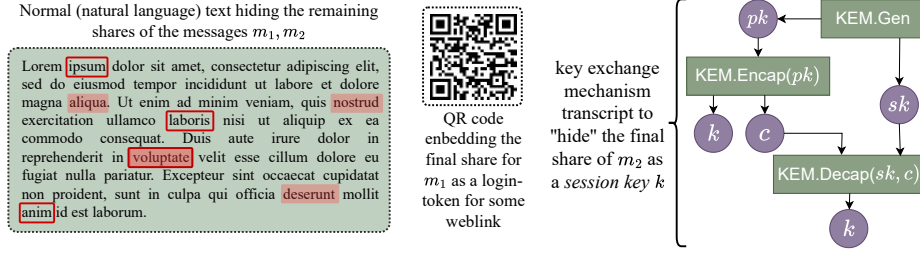


Fig. 1: PhantomCrypt at a glance. A natural-language cover text hides the remaining for two messages: m_1 (red boxes) and m_2 (gray highlight). Cryptographic protocols themselves become carriers of hidden information, ensuring both existence and content deniability.

The combination. IE and FBE are combinable because they rest on the same mechanism: secret sharing over $\mathbb{F}_p$. To make IE content-deniable, Sarah embeds m^* and ℓ decoy messages into an FBE container c_{FBE} ; the IE mechanism then derives shares from the cover text and computes a final share s_{new} encoding m^* . This share can be delivered in two ways: (a) as randomness inside an ordinary cryptographic protocol (a session key in a KEM exchange, a challenge in an authentication protocol³), in which case the cover text remains unmodified; or (b) steganographically embedded in the cover text (QR code or image), in which case the cover grows with each added message as in standalone FBE.

Robert, who shares the seed x_0 , reconstructs m^* by re-deriving shares and interpolating. Under coercion, he selects a different share set that reconstructs a plausible decoy. The deniability mechanism is itself deniable: an adversary without x_0 cannot tell that s_{new} served as a share, let alone which message it encodes. FBE provides CD, IE provides ED, and because both rest on the same algebraic foundation, the use of either mechanism is deniable.

3 Formal Definitions

We formalize content deniability, existence deniability, and second-order deniability through game-based definitions. All definitions are agnostic to the choice of building blocks; the specific construction PhantomCrypt is introduced in Section 5. Table 1 lists the principal symbols. For a string x , we write $|x|$ for its length, and $\text{poly}(\cdot)$ will denote any arbitrary but fixed univariate polynomial.

3.1 Scheme Syntax

Our scheme is a *symmetric plan-ahead* type deniable encryption (as opposed to ad-hoc or public-key schemes, whose construction we leave for future work).

³ Protocols whose nonces or randomness are not made visible to the opposite party (e.g., probabilistic encryption, certain signature schemes) are not suitable carriers for s_{new} .

Table 1: Notation: symbols and their meanings.

Symbol	Type	Meaning
λ	Integer	Security parameter
p	Prime	Defines $\mathbb{F}_p$
ℓ	Integer	Number of decoy messages
τ / T	String	Cover string (generic) / cover text (PhantomCrypt)
sk_{den}	Key	Deniable secret key (withheld under coercion)
sk_{std}	Key	Standard secret key (on-device, accessible)
dk_i	Key	i -th decoy decryption key
m^*, m_i	$\{0, 1\}^{\text{poly}(\lambda)}$	True message; i -th decoy message
k	Integer	Secret sharing threshold (PhantomCrypt)
SK_{ie}	Key tuple	IE secret (x_0, k) ; instantiates sk_{den}
sk_{kem}	Key	KEM decapsulation key; instantiates sk_{std}
$SK_{\text{FBE}, i}$	Key	i -th FBE key; instantiates dk_i
s_j, s_{new}	$\mathbb{F}_p$	j -th share $H(w_{i_j})$; transmitted share $P(x_{\text{new}})$
c_{FBE}	Vector	FBE ciphertext container
C_{kem}, C_1, C_2	Bytes	KEM ciphertext; AEAD-encrypted payloads
H	Hash	$\{0, 1\}^* \rightarrow \mathbb{F}_p$ (random oracle)

Compared to the classical three-algorithm syntax of [11], we separate three key roles: a deniable secret sk_{den} (withheld under coercion), a standard secret sk_{std} (accessible under coercion), and a set of per-message decoy keys $\{dk_i\}$. We likewise separate true-message encryption from decoy insertion.

Definition 1 (Deniable Encryption with Existential Hiding (DEEH)).
A DEEH scheme is a tuple $\Pi = (\text{Setup}, \text{Enc}, \text{AddDecoy}, \text{DecTrue}, \text{DecDecoy})$ of PPT algorithms:

- $\text{Setup}(\lambda, \tau, \ell) \rightarrow (\text{PP}, sk_{\text{den}}, sk_{\text{std}})$.
On input the security parameter λ , a cover string τ with $|\tau| = \text{poly}(\lambda)$, and a decoy budget $\ell \in \mathbb{N}$, outputs public parameters PP , a deniable secret sk_{den} (withheld under coercion), and a standard secret sk_{std} (accessible under coercion).
- $\text{Enc}(\text{PP}, sk_{\text{den}}, sk_{\text{std}}, m^*, \tau) \rightarrow (C, dk_0)$.
Encrypts the true message $m^ \in \{0, 1\}^{\text{poly}(\lambda)}$ into ciphertext C and returns dk_0 , whose decoy opening also yields m^* (so that dk_0 is structurally indistinguishable from a decoy key).*
- $\text{AddDecoy}(\text{PP}, sk_{\text{std}}, C, m_i) \rightarrow (C', dk_i)$.
Extends C by embedding a decoy message m_i and returns the updated ciphertext C' together with the corresponding decoy key dk_i ($1 \leq i \leq \ell$).
- $\text{DecTrue}(\text{PP}, sk_{\text{den}}, sk_{\text{std}}, C, \tau) \rightarrow m^*$ or $\perp$.
The honest recipient recovers the true message using the deniable secret.

- $\text{DecDecoy}(\text{PP}, sk_{\text{std}}, dk_i, C) \rightarrow m_i \text{ or } \perp$.
Recovers the decoy message associated with dk_i ; returns $\perp$ if dk_i was never produced on C .

Definition 2 (Correctness). A DEEH scheme is correct if, for all $\lambda \in \mathbb{N}$, all cover strings τ , all decoy budgets ℓ , all true messages m^* and decoy messages $\{m_i\}_{i=1}^\ell$ with $m^*, m_1, \dots, m_\ell$ pairwise distinct, the following holds with probability 1 in an honest execution. Let

$$\begin{aligned} (\text{PP}, sk_{\text{den}}, sk_{\text{std}}) &\leftarrow \text{Setup}(\lambda, \tau, \ell), \\ (C_0, dk_0) &\leftarrow \text{Enc}(\text{PP}, sk_{\text{den}}, sk_{\text{std}}, m^*, \tau), \end{aligned}$$

and iteratively $(C_i, dk_i) \leftarrow \text{AddDecoy}(\text{PP}, sk_{\text{std}}, C_{i-1}, m_i)$ for $i = 1, \dots, \ell$. Then (i) $\text{DecTrue}(\text{PP}, sk_{\text{den}}, sk_{\text{std}}, C_\ell, \tau) = m^*$; (ii) $\text{DecDecoy}(\text{PP}, sk_{\text{std}}, dk_0, C_\ell) = m^*$; and (iii) $\text{DecDecoy}(\text{PP}, sk_{\text{std}}, dk_i, C_\ell) = m_i$ for every $i \in \{1, \dots, \ell\}$.

3.2 Content Deniability (CD)

Content deniability captures the property that an adversary who coerces the honest party into disclosing decryption keys cannot identify which message was originally designated as the true one among the decoys. We formalize this via an indistinguishability game, then introduce a complementary asymptotic notion capturing the adversary's *residual uncertainty* about how many messages are hidden.

Definition 3 (CD Security Game). For a DEEH scheme Π with decoy budget ℓ and adversary $\mathcal{A}$, the game $\text{Game}_{\Pi, \mathcal{A}}^{\text{CD}}(\lambda)$ proceeds as follows:

1. $\mathcal{A}$ submits a cover string τ with $|\tau| = \text{poly}(\lambda)$ and $\ell + 1$ pairwise distinct messages $m_0, m_1, \dots, m_\ell$.
2. The challenger runs $(\text{PP}, sk_{\text{den}}, sk_{\text{std}}) \leftarrow \text{Setup}(\lambda, \tau, \ell)$.
3. The challenger samples $b^* \xleftarrow{\$} \{0, \dots, \ell\}$, sets $m^* := m_{b^*}$, and runs $(C, dk_{b^*}) \leftarrow \text{Enc}(\text{PP}, sk_{\text{den}}, sk_{\text{std}}, m^*, \tau)$. Then, for every $i \in \{0, \dots, \ell\} \setminus \{b^*\}$, the challenger runs $(C, dk_i) \leftarrow \text{AddDecoy}(\text{PP}, sk_{\text{std}}, C, m_i)$.
4. The challenger hands $\mathcal{A}$ the tuple $(\text{PP}, \tau, C, sk_{\text{std}}, \{dk_i\}_{i=0}^\ell)$. The deniable secret sk_{den} is withheld.
5. $\mathcal{A}$ outputs a guess $b' \in \{0, \dots, \ell\}$; $\mathcal{A}$ wins iff $b' = b^*$.

Define $\text{Adv}_{\Pi, \mathcal{A}}^{\text{CD}}(\lambda) = |\Pr[b' = b^*] - 1/(\ell + 1)|$.

Definition 4 (Content Deniability). A DEEH scheme Π is ε_{CD} -content-deniable if $\text{Adv}_{\Pi, \mathcal{A}}^{\text{CD}}(\lambda) \leq \varepsilon_{\text{CD}}(\lambda)$ for every PPT adversary $\mathcal{A}$. We say Π is content-deniable if $\varepsilon_{\text{CD}}(\lambda) = \text{negl}(\lambda)$.

A complementary (t, q) -framework. Definition 3 fixes the decoy pool in advance to size $\ell+1$. It is useful to complement it with an open-world notion that captures the adversary’s residual uncertainty after coercion. We call Π (t, q) -deniable if every adversary with time complexity $\leq t(\lambda)$ and $\leq q$ secret-key queries to the honest party cannot rule out that a further, yet-undiscovered message is embedded in the challenger’s ciphertext C . Trivially $q \leq t$; the interesting regime is $q < \ell$, in which the adversary has consumed its query budget before exhausting the decoy pool. The two notions are related but not equivalent: CD security asks whether the adversary can *identify* the true message among a known pool, whereas (t, q) -deniability asks whether the adversary can *rule out further* hidden messages.

Example 1 (Canetti et al.). The sender-deniable PKE of [11] admits exactly one alternative opening. An adversary issuing a single key query receives the unique decoy opening, after which no further consistent message can be plausibly exhibited by the sender. Hence the scheme is $(\text{poly}(\lambda), 1)$ -deniable but not $(\text{poly}(\lambda), q)$ -deniable for any $q \geq 2$.

Example 2 (FBE). FBE [1] embeds $\ell + 1$ messages via an underdetermined linear system over $\mathbb{F}_p$. An adversary holding $q \leq \ell$ keys recovers q messages; the remaining $\ell + 1 - q$ slots are information-theoretically consistent with arbitrary field elements. FBE is therefore (∞, ℓ) -deniable. Residual uncertainty degrades to zero only once the adversary obtains all $\ell + 1$ keys.

For PhantomCrypt specifically (Section 5), the set of candidate decoy openings is a subset of k -element position tuples drawn from the L words of the cover text τ . The space has size $\binom{L}{k} = \Theta(2^L)$ for $k = \Theta(L)$, so any polynomial-time adversary fails to exhaust it: PhantomCrypt is $(\text{poly}(\lambda), \text{poly}(\lambda))$ -deniable in the sense above. This super-polynomial residual set is what prevents the attacker from concluding “there is no further hidden message” even after arbitrary polynomial effort.

3.3 Existence Deniability (ED)

Existence deniability captures the property that a transmitted document D is, in parts, indistinguishable from traffic produced by an innocent conventional (i.e., not deniable) cryptographic protocol, or transmission of randomly looking data (like an authentication token in a QR code). To this end, we introduce the concept of a *reference distribution* that captures the distribution of documents.

Definition 5 (Reference Distribution). A reference distribution $\mathcal{D}_{\text{ref}}$ is a family of efficiently samplable distributions, parameterized by a string ω .

To make this concrete, consider $\tau' \leftarrow \mathcal{D}_{\text{ref}}(\tau \parallel \text{PP})$, where τ is a cover text and PP are the system parameters of a PQC-KEM scheme, concatenated in an efficiently decodable form (denoted $\parallel$). This sampling emits τ *extended* by the transcript of a KEM session-key agreement under parameters PP. The crucial

point: an attacker cannot tell whether the KEM session key was the IE final share s_{new} or an independent uniform string. Definition 6 formalizes this by letting the challenger decide whether to embed the secret message in the document, and letting the attacker guess.

Definition 6 (ED Security Game). For a DEEH scheme Π and adversary $\mathcal{A}$, the game $\text{Game}_{\Pi, \mathcal{A}}^{\text{ED}}(\lambda)$ proceeds as follows:

1. $\mathcal{A}$ submits a cover string τ . We require $|\tau| \geq \kappa(\lambda)$ for some public polynomial κ ; for *PhantomCrypt*, τ must consist of at least $k + 1$ distinct words (so that the PRNG can select $k - 1$ share positions and a fresh evaluation point $x_{\text{new}} \notin \{0, x_1, \dots, x_{k-1}\}$ without collision). Shorter cover strings trivially fail ED by leaking the mechanism through structural impossibility.
2. The challenger runs $(\text{PP}, sk_{\text{den}}, sk_{\text{std}}) \leftarrow \text{Setup}(\lambda, \tau, \ell)$ and samples $b \xleftarrow{\$} \{0, 1\}$.
3. $\mathcal{A}$ submits a message $m^* \leftarrow \mathcal{D}_{\text{ref}}(\tau)$.
4. If $b = 1$: the challenger runs $(C, dk_0) \leftarrow \text{Enc}(\text{PP}, sk_{\text{den}}, sk_{\text{std}}, m^*, \tau)$.
If $b = 0$: the challenger samples $C \leftarrow \mathcal{D}_{\text{ref}}(\tau \parallel \text{PP})$.
In either case the challenger returns (τ, C) to $\mathcal{A}$.
5. $\mathcal{A}$ outputs a guess $b' \in \{0, 1\}$.

Define $\text{Adv}_{\Pi, \mathcal{A}}^{\text{ED}}(\lambda) = |\Pr[b' = b] - 1/2|$.

Definition 7 (Existence Deniability). A DEEH scheme is ε_{ED} -existence-deniable w.r.t. $\mathcal{D}_{\text{ref}}$ if $\text{Adv}_{\Pi, \mathcal{A}}^{\text{ED}}(\lambda) \leq \varepsilon_{\text{ED}}(\lambda)$ for every PPT $\mathcal{A}$. We say Π is existence-deniable if $\varepsilon_{\text{ED}}(\lambda) = \text{negl}(\lambda)$.

Remark 1. $\mathcal{D}_{\text{ref}}$ is an explicit parameter: different instantiations arise from different baselines (e.g., PQC-KEM hybrid encryption, TLS records). The cover string τ can be a natural-language text, an image, or any carrier. In both branches of the ED game, the adversary sees the same τ ; the challenge is solely whether the ciphertext was produced by *PhantomCrypt* or by the reference protocol.

3.4 Second-Order Deniability

Second-order deniability requires that the use of a deniability mechanism is itself deniable: an adversary can neither detect the mechanism from the ciphertext nor identify the true message under coercion. Security is here, unlike for ED or CD, understood in a passive sense, i.e., we will let the honest party (challenger) set up the system, choose the cover text, define the sensitive message m^* and choose its decoys $m_1, \dots, m_\ell$. This is to avoid trivial winning strategies for the attacker, like prescribing a cover text that is simply too short to take up any secret (length of τ much smaller than the length of the message to be embedded), or enforce a cover text that is a repetition of only one word, which would imply known values for each s_i (although not necessarily imply a likewise knowledge about the x -values, thus retaining some security, but nonetheless leaking out information too).

Definition 8 (Second-Order Deniability; informal). *A DEEH scheme Π achieves second-order deniability w.r.t. $\mathcal{D}_{\text{ref}}$ if no PPT adversary can, with non-negligible probability, determine, from the transmitted ciphertext and any information obtained under coercion (excluding sk_{den}), that a deniability mechanism was used to conceal a particular message.*

Proposition 1. *A DEEH scheme achieves second-order deniability if and only if it is both ε_{CD} -content-deniable (Definition 4) and ε_{ED} -existence-deniable (Definition 7) with both advantages negligible.*

Proof. Let us define a scenario like in the CD and ED games, only with the attacker put into the role of a passive cryptanalyst, with (physical power of coercion) over the challenger who prepared the ciphertext. We model the attacker's coercive power as oracle access to the challenger $\mathcal{C}$, letting the attacker ask for encryptions and decryptions of a given message, but without access to the secret knowledge, i.e., keys, of $\mathcal{C}$:

1. The challenger chooses a cover text τ and samples pairwise distinct messages $m_0, \dots, m_\ell \leftarrow \mathcal{D}_{\text{ref}}(\tau)$. It runs $(\text{PP}, sk_{\text{den}}, sk_{\text{std}}) \leftarrow \text{Setup}(\lambda, \tau, \ell)$.
2. The challenger samples $b^* \xleftarrow{\$} \{0, \dots, \ell\}$, sets $m^* := m_{b^*}$, and runs $(C, dk_{b^*}) \leftarrow \text{Enc}(\text{PP}, sk_{\text{den}}, sk_{\text{std}}, m^*, \tau)$; for every $i \in \{0, \dots, \ell\} \setminus \{b^*\}$ it runs $(C, dk_i) \leftarrow \text{AddDecoy}(\text{PP}, sk_{\text{std}}, C, m_i)$.
3. The adversary receives the full message list $m_0, \dots, m_\ell$ (but not the index b^*).
4. Under coercion, the challenger reveals sk_{std} and all $\ell + 1$ decoy keys $\{dk_i\}_{i=0}^\ell$ but withholds sk_{den} .

($\Leftarrow$) ED prevents detection of the mechanism; CD prevents identification of the true message under coercion. Together, the mechanism's use is deniable. ($\Rightarrow$) Failure of ED allows detection; failure of CD allows identification, confirming the mechanism was used. $\square$

Lemma 1 (Two-stage composition). *Let Π be ε_1 -ED-secure and ε_2 -CD-secure. A two-stage adversary $\mathcal{A}$ that first observes (τ, C) and outputs a detection bit d , then (if $d = 1$) receives $(sk_{\text{std}}, \{dk_i\})$ and outputs b'_c , satisfies:*

$$\Pr[\text{detect} \wedge \text{identify}] \leq \left(\frac{1}{2} + \varepsilon_1\right) \left(\frac{1}{\ell+1} + \varepsilon_2\right).$$

If $\varepsilon_1, \varepsilon_2 = \text{negl}(\lambda)$, this is at most $\frac{1}{2(\ell+1)} + \text{negl}(\lambda)$: correctly detecting deniable encryption and identifying the true message is only negligibly better than guessing among $2(\ell+1)$ possibilities.

Proof. Condition on the event that the challenge ciphertext is generated by Π (i.e., $b = 1$ in the ED game). In the detection phase, $\mathcal{A}$'s view is the ED game, so for an adversary applying the ED-optimal distinguisher, $\Pr[d = 1 \mid b = 1] \leq$

$1/2 + \varepsilon_1$. Conditioned on $d = 1$, $\mathcal{A}$'s additional input is exactly the CD challenge of Definition 3, so $\Pr[b'_c = b_c \mid d = 1, b = 1] \leq 1/(\ell + 1) + \varepsilon_2$. Chaining:

$$\begin{aligned} \Pr[\text{detect} \wedge \text{identify} \mid b = 1] &= \Pr[d=1 \mid b=1] \cdot \Pr[b'_c=b_c \mid d=1, b=1] \\ &\leq \left(\frac{1}{2} + \varepsilon_1\right) \left(\frac{1}{\ell+1} + \varepsilon_2\right). \end{aligned}$$

Expanding and dropping products of negligibles gives $\frac{1}{2(\ell+1)} + \text{negl}(\lambda)$. $\square$

3.5 Independence of CD and ED

Lemma 2 (Independence). *Content deniability and existence deniability are logically independent.*

Proof. $CD \not\Rightarrow ED$. FBE [1] embeds multiple messages in a single ciphertext via underdetermined linear systems, achieving information-theoretic CD. However, its ciphertext is a raw vector of field elements, trivially distinguishable from standard protocol traffic: $\text{Adv}^{\text{ED}} = 1/2 - \text{negl}(\lambda)$.

$ED \not\Rightarrow CD$. IE [2] embeds a secret message as the constant term of a polynomial interpolated through shares derived from an unmodified cover text, with the transmitted share carried in a standard hybrid encryption envelope. IE achieves ED but embeds only a single message with no prepared decoy keys. In the CD game with $\ell \geq 1$, the adversary identifies m^* with probability 1. $\square$

Remark 2 (IE does not provide CD). In IE, a party could construct a different polynomial through the same shares, yielding a different constant term. However, the CD game gives the adversary sk_{std} and all decoy keys. IE has no decoy keys: an ad hoc alternative does not constitute a decoy key in Definition 1.

4 Related Work

We place related work after the formal definitions so that existing schemes can be classified by which of CD and ED they achieve.

Content deniability schemes. Canetti et al. [11] introduced deniable encryption; subsequent work [10] achieved full bi-deniability via iO. An and Zhang [5] introduced dual-deniable PKE; An et al. [4] achieved leakage-resilient deniable encryption. FBE [1] supports arbitrary decoys via underdetermined linear systems, achieving information-theoretic CD. Honey Encryption [27] renders incorrect-key decryptions plausible. Hidden volumes (VeraCrypt [32], Shufflecake [7]) provide CD at the storage level. Canetti et al. [9] introduced deniable secret sharing, exploiting the same algebraic structure as FBE. All produce ciphertexts distinguishable from standard traffic, failing ED.

Existence deniability and steganography. Traditional steganography [15, 26, 33] and linguistic steganography [16, 17] embed hidden messages in cover media. Provably secure generative steganography: Kaptchuk et al. [28] via GPT-2; De Witt et al. [35] via minimum entropy coupling; Ding et al. [21] via distribution copies. IE [2] derives shares from an unmodified cover text within a hybrid encryption envelope. These achieve ED but not CD.

Anamorphic encryption. Persiano et al. [30] introduced anamorphic encryption. Banfi et al. [8] proposed robust anamorphic encryption; Catalano et al. [14] extended it to hybrid encryption, IBE, and FHE; Catalano et al. [13] proved generic impossibility for stateless black-box anamorphic encryption with large message spaces; Persiano et al. [31] introduced public-key anamorphism; Choi et al. [18] unified receiver notions; Deo and Libert [20] gave the first lattice-based construction; Carnemolla et al. [12] and Dodis and Goldin [22] constructed anamorphic-resistant PKE. Anamorphic encryption achieves a form of ED, but normal and covert decryption are structurally asymmetric, so an adversary receiving all keys can identify the “normal” mode. Moreover, it supports only two messages, and the impossibility results of [13] imply capacity limits not applying to FBE.

Obfuscated channels and censorship resistance. Protocol obfuscation (obfs4 [6], ScrambleSuit [34]) disguises Tor traffic. Fenske and Johnson [23, 24] formalized Fully Encrypted Protocols; Günther et al. [25] formalized obfuscated key exchange with ML-KEM. Wu et al. [36] documented Great Firewall detection of fully encrypted traffic. These are network-layer ED mechanisms that do not address CD.

At-compromise security. Albrecht et al. [3] introduced alert blindness from ethnographic fieldwork during Kenya’s 2024 protests; Collins et al. [19] found that cryptographic deniability is never invoked by defendants in Swiss courts.

Position of PhantomCrypt. PhantomCrypt is the first construction achieving CD and ED simultaneously with formal game-based definitions and explicit security reductions, providing $\ell + 1$ structurally symmetric decryptions and network-level indistinguishability from standard hybrid encryption.

5 PhantomCrypt Construction

PhantomCrypt instantiates the DEEH syntax (Definition 1) as follows: the cover string τ is a natural-language text $T = (w_1, \dots, w_L)$; $\mathcal{M} = \mathbb{F}_p$; sk_{den} is the IE secret $SK_{\text{ie}} = (x_0, k)$, pre-shared via IE [2]; sk_{std} is sk_{kem} ; each dk_i is an FBE key $SK_{\text{FBE},i}$; and $\mathcal{D}_{\text{ref}}$ is the output of a standard PQC-KEM hybrid encryption.

5.1 Building Blocks

Invisible Encryption (IE) [2]. Given a cover text $T = (w_1, \dots, w_L)$ and a secret seed x_0 , IE deterministically selects $k - 1$ words from T using a PRNG seeded

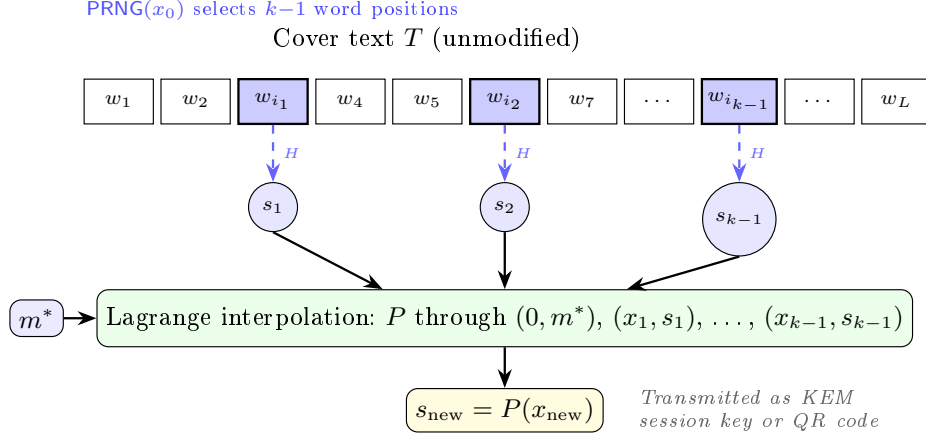


Fig. 2: IE share derivation. A PRNG seeded by x_0 selects $k-1$ words from cover text T . Each word is hashed to produce a share $s_j = H(w_{i_j})$. A polynomial P is interpolated through these shares and $(0, m^*)$. The transmitted share $s_{\text{new}} = P(x_{\text{new}})$ is pseudorandom and embeddable in any channel admitting random payloads. The cover text is never modified.

by x_0 . Each selected word w_{i_j} produces a share $s_j = H(w_{i_j}) \in \mathbb{F}_p$ at abscissa x_j . A degree- $(k-1)$ polynomial P is interpolated through Equation (1), and the transmitted share $s_{\text{new}} = P(x_{\text{new}})$ is disguised as a session key within a standard KEM exchange. The cover text is *never modified*: shares are derived from it, not embedded into it. Figure 2 illustrates this process.

Provides: ED. The transmitted ciphertext (KEM ciphertext + AEAD payload) is syntactically and distributionally consistent with standard hybrid encryption. *Does not provide*: CD. The polynomial P is uniquely determined by x_0 and s_{new} . An adversary who obtains x_0 recovers m^* with certainty; there are no prepared decoy keys.

False-Bottom Encryption (FBE) [1]. FBE embeds multiple messages within a single ciphertext container $c_{\text{FBE}} \in \mathbb{F}_p^N$ using underdetermined linear equations. Each message m_i is associated with a decryption key $\text{SK}_i = (I_{c,i}, I_{r,i}, \text{pos}_i)$ specifying which container elements and key-base elements to combine. The system of equations is underdetermined, ensuring that the container is consistent with all $\ell + 1$ messages simultaneously. All keys are structurally symmetric: no key is distinguishable as the “true” one from the ciphertext and keys alone. *Provides*: CD. Given c_{FBE} and all $\ell + 1$ keys, an adversary cannot determine which message was designated as true. *Does not provide*: ED. The container consists of raw field elements with no resemblance to standard protocol traffic. Figure 3 illustrates the process flow. IE takes the cover text T and a message m , derives $k-1$ shares from selected words, and computes the final share $s_{\text{new}} = P(x_{\text{new}})$. A protocol Π is selected from a suite of ordinary cryptographic protocols (KEM

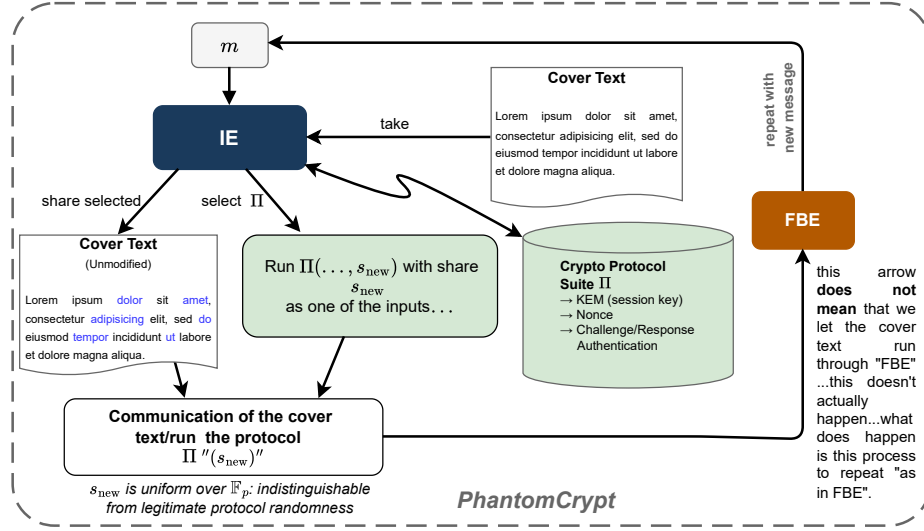


Fig. 3: PhantomCrypt process flow.

exchange, challenge-response authentication, etc.), and s_{new} is embedded as its randomness (nonce, session key, or challenge). Both the cover text and the protocol transcript are communicated to the receiver. Since s_{new} is uniform over $\mathbb{F}_p$, it is indistinguishable from the legitimate randomness of Π (ED). The FBE loop adds decoy messages iteratively, each with a structurally symmetric decryption key (CD).

5.2 Composition

FBE embeds m^* and ℓ decoys into c_{FBE} with symmetric keys (providing CD). IE independently derives shares from the unmodified cover text and computes s_{new} (providing ED). A PQC-KEM establishes a session key K_{session} that encrypts two AEAD payloads:

$$C_1 = \text{AE.Enc}(K_{\text{session}}, s_{\text{new}}, \text{IV}_1), \quad C_2 = \text{AE.Enc}(K_{\text{session}}, c_{\text{FBE}}, \text{IV}_2).$$

The full ciphertext is $C = (C_{\text{kem}}, C_1, C_2, \text{IV}_1, \text{IV}_2)$.

True decryption requires $\text{SK}_{\text{ie}} = (x_0, k)$, sk_{kem} , and the cover text T : the receiver decapsulates the KEM, decrypts C_1 to recover s_{new} , re-derives the $k-1$ shares from T using the pre-shared seed x_0 , and reconstructs $m^* = P(0)$ via Lagrange interpolation.

Decoy decryption requires sk_{kem} and an FBE key $\text{SK}_{\text{FBE},i}$: the receiver decapsulates the KEM, decrypts C_2 to recover c_{FBE} , and applies FBE decryption. Since m^* is also embedded in FBE, decoy decryption with the FBE key for m^* also recovers m^* but the key is structurally indistinguishable from any other FBE key, providing no information about which message is true.

Remark 3 (PhantomCrypt coercion model). In the CD game (Definition 3), PhantomCrypt instantiates $sk_{\text{std}} = sk_{\text{kem}}$ and $\{dk_i\} = \{SK_{\text{FBE},i}\}$. With these, the adversary decrypts C_1 (recovering s_{new}) and C_2 (recovering c_{FBE}), then uses the FBE keys to recover all $\ell + 1$ messages. The withheld deniable key $SK_{\text{ie}} = (x_0, k)$ is pre-shared per the IE protocol [2]; x_0 is a 256-bit seed representable as a 24-word mnemonic, making it memorizable and destroyable before seizure.

Remark 4 (Transmission channels for s_{new}). The share s_{new} is pseudorandom and can be embedded in any channel admitting random payloads. In the primary instantiation it is carried within the AEAD payload of the PQC-KEM ciphertext. Alternatively, s_{new} can be embedded in a QR code, with $\mathcal{D}_{\text{ref}}$ redefined accordingly. The choice of channel affects contextual plausibility but not the cryptographic bounds.

Why composition is natural. Information-theoretic CD requires algebraic structure admitting multiple consistent decryptions, while ED requires ciphertexts matching a standard protocol distribution. These goals are in tension: raw algebraic objects do not resemble protocol traffic, and standard ciphertexts typically admit unique decryptions. PhantomCrypt resolves this by composing FBE (algebraic structure for CD) with IE+KEM (protocol-conformant format for ED). Alternative architectures exist (e.g., a KEM with Canetti-style deniable encryption [11]), but PhantomCrypt uniquely achieves information-theoretic CD ($\varepsilon_{\text{CD}} = 0$).

6 Security Analysis

We establish PhantomCrypt’s CD and ED properties via reductions to its building blocks.

6.1 Existence Deniability

Theorem 1 (ED Security). *Let $\mathcal{D}_{\text{ref}}(\tau)$ be the distribution of ciphertexts from standard hybrid PQC-KEM encryption of a random payload over cover string τ . If the KEM is IND-CCA secure with advantage ε_{KEM} and the AEAD is IND-CPA secure with advantage $\varepsilon_{\text{AEAD}}$, then $\varepsilon_{\text{ED}} \leq \varepsilon_{\text{KEM}} + 2\varepsilon_{\text{AEAD}}$.*

Proof. Game 0: The real ED game. The challenger runs PhantomCrypt: encapsulates to obtain $(C_{\text{kem}}, K_{\text{session}})$, encrypts $C_1 = \text{AE.Enc}(K_{\text{session}}, s_{\text{new}}, IV_1)$ and $C_2 = \text{AE.Enc}(K_{\text{session}}, c_{\text{FBE}}, IV_2)$.

Game 1: Replace K_{session} with $\tilde{K} \xleftarrow{\$} \{0, 1\}^{256}$. By IND-CCA of the KEM, $|\Pr[G0] - \Pr[G1]| \leq \varepsilon_{\text{KEM}}$.

Game 2: Replace C_1 with encryption of random r_1 . By IND-CPA of AEAD under $\tilde{K}$, $|\Pr[G1] - \Pr[G2]| \leq \varepsilon_{\text{AEAD}}$.

Game 3: Replace C_2 with encryption of random r_2 . Similarly, $|\Pr[G2] - \Pr[G3]| \leq \varepsilon_{\text{AEAD}}$.

In Game 3, the ciphertext is a KEM ciphertext followed by two AEAD encryptions of random payloads under a random key: exactly $\mathcal{D}_{\text{ref}}(\tau)$. The cover text T is identical in all games (IE never modifies it; the adversary's submitted τ is used throughout). By the triangle inequality: $\varepsilon_{\text{ED}} \leq \varepsilon_{\text{KEM}} + 2\varepsilon_{\text{AEAD}}$. $\square$

6.2 Content Deniability

Lemma 3 (Uniformity of IE share). *In the random oracle model, for any fixed seed x_0 , cover text T , and threshold $k \geq 3$, the transmitted share $s_{\text{new}} = P(x_{\text{new}})$ is uniformly distributed over $\mathbb{F}_p$ regardless of the message $m = P(0)$.*

Proof. By Lagrange interpolation, $s_{\text{new}} = m \cdot L_0(x_{\text{new}}) + \sum_{j=1}^{k-1} s_j \cdot L_j(x_{\text{new}})$, where L_j are basis polynomials for abscissae $\{0, x_1, \dots, x_{k-1}\}$ and $s_j = H(w_{i_j})$. In the random oracle model, $s_1, \dots, s_{k-1}$ are mutually independent and uniform over $\mathbb{F}_p$. The IE protocol ensures $x_{\text{new}} \notin \{0, x_1, \dots, x_{k-1}\}$ (collision avoidance), so at least one $L_j(x_{\text{new}}) \neq 0$ for $j \geq 1$. Hence s_{new} is a nontrivial affine function of uniform values, and therefore uniform over $\mathbb{F}_p$ for any fixed m . In particular, for any $m, m' \in \mathbb{F}_p$, the distributions of s_{new} conditioned on $P(0) = m$ and $P(0) = m'$ are both uniform over $\mathbb{F}_p$ and hence identical. $\square$

Theorem 2 (CD Security). *In the random oracle model, PhantomCrypt is ε_{CD} -content-deniable with $\varepsilon_{\text{CD}} = 0$. Even a computationally unbounded adversary in the CD game (Definition 3) cannot identify the true message with advantage better than $1/(\ell + 1)$.*

Proof. In the CD game, the adversary receives C , sk_{kem} , and all $\ell + 1$ FBE keys $\{\text{SK}_{\text{FBE},i}\}_{i=0}^{\ell}$, but not the IE key $\text{SK}_{\text{ie}} = (x_0, k)$. Using sk_{kem} , the adversary decrypts C_1 to obtain $s_{\text{new}} \in \mathbb{F}_p$ and C_2 to obtain c_{FBE} . Using the FBE keys, the adversary recovers all $\ell + 1$ messages. We show the adversary's view is identically distributed for all values of b^* .

FBE symmetry. All $\ell + 1$ messages are embedded in c_{FBE} with structurally symmetric keys. By the information-theoretic security of FBE [1], the joint distribution of $(c_{\text{FBE}}, \{\text{SK}_{\text{FBE},i}\}_{i=0}^{\ell})$ is identical for every permutation of which message is designated “true.” Therefore, the FBE component reveals no information about b^* .

IE component. By Lemma 3, s_{new} is uniform over $\mathbb{F}_p$ regardless of which message m_{b^*} was placed at $P(0)$. The adversary does not possess x_0 and hence cannot recover the abscissae, shares, or x_{new} . Consequently, s_{new} is statistically independent of b^* .

Independence of layers. The IE layer (producing s_{new}) and the FBE layer (producing c_{FBE} and keys) use independent randomness. Therefore, the joint view

$$(s_{\text{new}}, c_{\text{FBE}}, \{\text{SK}_{\text{FBE},i}\}_{i=0}^{\ell}, T)$$

is identically distributed for all values of b^* . Consequently, $\Pr[b' = b^*] = 1/(\ell + 1)$ for any adversary, and $\varepsilon_{\text{CD}} = 0$. $\square$

6.3 Second-Order Deniability

Corollary 1 (PhantomCrypt achieves second-order deniability). *When instantiated with an IND-CCA secure PQC-KEM and an IND-CPA secure AEAD scheme, PhantomCrypt achieves second-order deniability with $\varepsilon_{\text{ED}} \leq \varepsilon_{\text{KEM}} + 2\varepsilon_{\text{AEAD}} = \text{negl}(\lambda)$ (Theorem 1) and $\varepsilon_{\text{CD}} = 0$ in the random oracle model (Theorem 2). By Lemma 1, a two-stage adversary who first observes the network and then coerces for keys succeeds with probability at most $\frac{1}{2(\ell+1)} + \text{negl}(\lambda)$.*

6.4 Post-Quantum Security

PhantomCrypt’s security against quantum adversaries rests on two pillars. FBE’s content deniability relies on linear algebra over $\mathbb{F}_p$ and information-theoretic secret sharing, which hold regardless of computational model. The ED property relies on KEM indistinguishability and session-key pseudorandomness; when instantiated with ML-KEM-768 (IND-CCA under Module-LWE [29]) and H modeled as a quantum random oracle, these resist quantum adversaries. The FBE component of CD is information-theoretic, the IE component extends naturally to the QROM (since the shares remain uniform under quantum random oracle queries), and ε_{ED} is negligible under PQC assumptions. A formal end-to-end QROM proof remains an open problem.

7 Multi-Message Security

Single-message deniability does not automatically extend to multi-message settings. We identify a concrete vulnerability and propose mitigations.

Lemma 4 (Key reuse vulnerability). *If IE encrypts distinct messages m_1, m_2 with the same seed x_0 , cover text T , and threshold k , then $m_1 - m_2 \pmod{p}$ is recoverable from the transmitted shares $s_{\text{new},1}$ and $s_{\text{new},2}$.*

Proof. Both messages share $k - 1$ points: $P_1(x_j) = P_2(x_j) = s_j$ for $j \in [k - 1]$, since the same seed, cover text, and parameters determine the same word selections and hashes. The difference polynomial $D(x) = P_1(x) - P_2(x)$ has degree $\leq k - 1$ and vanishes at $k - 1$ known points $\{x_j\}$. Hence $D(x) = c \cdot \prod_{j=1}^{k-1} (x - x_j)$ for some constant c . Since $D(x_{\text{new}}) = s_{\text{new},1} - s_{\text{new},2}$ is known to the adversary (who observes both transmitted shares), c is determined, and $D(0) = m_1 - m_2$ is recoverable. $\square$

Mitigation 1: Seed ratcheting. After each message, evolve the seed: $x_0^{(i+1)} \leftarrow H(x_0^{(i)} \| C^{(i)})$. Under the random oracle model, each derived seed is computationally indistinguishable from random, ensuring independent word selections and shares per message.

Theorem 3 (Seed ratcheting security). *Under seed ratcheting with random oracle H , for any polynomial-length message sequence $\{m_i\}$ and any PPT adversary observing all ciphertexts: (i) each message uses computationally independent IE parameters; (ii) the ED and CD advantages per message remain $\varepsilon_{\text{ED}} + \text{negl}(\lambda)$ and $1/(\ell + 1) + \text{negl}(\lambda)$ respectively; (iii) the difference attack of Lemma 4 is mitigated.*

Proof. We proceed by hybrid argument. Replace $x_0^{(i+1)} = H(x_0^{(i)} \| C^{(i)})$ with a truly random value $\tilde{x}_0^{(i+1)} \xleftarrow{\$} \{0, 1\}^\lambda$. Any adversary distinguishing these hybrids can be used to distinguish a random oracle output from random, contradicting the random oracle assumption. With independent seeds, the word selections and shares for each message are independent, eliminating the shared points that enable the difference attack. The per-message ED and CD guarantees follow from the single-message analysis applied independently to each message. $\square$

Additional mitigations. Cover text rotation (using distinct T_i per message) breaks the correlation independently of seed ratcheting. Ephemeral KEM keys (fresh $(pk_{\text{kem}}, sk_{\text{kem}})$ per session) provide forward secrecy. Combining all three techniques provides defense-in-depth.

8 Implementation and Evaluation

We implemented PhantomCrypt in Python, with the source code available at <https://anonymous.4open.science/r/PhantomCrypt-D532>. The implementation uses the prime $p = 2^{256} - 2^{32} - 977$ to define the finite field $\mathbb{F}_p$, SHA3-256 as the hash function H , ML-KEM-768 (FIPS 203 [29]) for the post-quantum KEM, and AES-256-GCM for symmetric encryption. All benchmarks were performed on an Intel Core i5-1135G7 running at 2.42 GHz with 16.0 GB of installed RAM (15.7 GB usable) under a 64-bit operating system and Python 3.10. Averaged over 100 runs, encryption of a 32-byte message using parameters $k = 5$ and $\ell = 3$ takes approximately 7.4 milliseconds, true decryption 7.0 milliseconds, and decoy decryption only 0.7 milliseconds, as the latter bypasses polynomial reconstruction. The total communication overhead for three decoys is 1464 bytes, dominated by the ML-KEM-768 ciphertext at 1088 bytes; importantly, this overhead does not scale with message size.

Application scenarios. Activists embed GPS coordinates within an academic paper (cover text), with five decoy messages for plausible cover. A whistleblower embeds a 32-byte document hash within a 2000-word article; under coercion, a decoy FBE key produces “Meeting confirmed.”

8.1 Kerckhoffs’ Principle and Scope of ED

PhantomCrypt’s security rests on keys, not algorithm secrecy. The key material is partitioned: the KEM key sk_{kem} is stored on the device and accessible under

coercion, while the IE secret $SK_{ie} = (x_0, k)$ is pre-shared, memorizable (a 256-bit seed representable as a 24-word mnemonic), and destroyable. CD security holds even if sk_{kem} and all FBE keys are compromised, provided SK_{ie} remains secret (Theorem 2).

PhantomCrypt is consistent with Kerckhoffs’ principle: an adversary who has read this paper and knows PhantomCrypt exists still cannot distinguish a PhantomCrypt ciphertext from standard PQC-KEM hybrid encryption (Theorem 1). The adversary knows that deniability mechanisms *could* be in use, but cannot determine whether any particular ciphertext employs one. A distinctive feature is that PhantomCrypt uses standard encryption as its own steganographic cover: the hybrid PQC-KEM ciphertext serves a dual role, functioning as a legitimate encryption under coercion (decrypted via any FBE key to yield a plausible message) while simultaneously carrying shares of the true message. The standard encryption is “degraded” to a decoy while the true communication hides within its structure.

Our ED claim is about network-level indistinguishability. We do *not* claim that PhantomCrypt hides itself from software inspection on a seized device. If the adversary discovers PhantomCrypt software (Stage 2 of Section 1.1), second-order deniability no longer holds, since the adversary has determined that a deniability mechanism is in use. The scheme degrades gracefully to first-order deniability (CD alone), which remains intact by Theorem 2.

9 Conclusion

We introduced second-order deniability *the property that the use of a deniability mechanism is itself deniable* and proved it achievable via the conjunction of content deniability and existence deniability, two properties we showed to be logically independent. The core insight is that both properties can be realized by the same algebraic primitive: threshold secret sharing via Lagrange interpolation over finite fields. PhantomCrypt composes FBE (for content deniability) with IE (for existence deniability) within a hybrid PQC-KEM envelope, and we proved that this composition preserves both security properties: ED via hybrid reductions to KEM and AEAD security, and CD via an information-theoretic argument in the random oracle model.

A distinctive feature of the construction is that standard encryption serves as its own steganographic cover. The hybrid PQC-KEM ciphertext simultaneously functions as a legitimate encrypted session (under coercion, decryptable via any FBE key to yield a plausible message) and as a carrier for the true message’s shares. This “degradation” of standard encryption to a decoy, while the true communication hides within, is the mechanism by which second-order deniability operates under Kerckhoffs’ principle.

Open problems include: (i) formally extending the CD argument to the QROM; (ii) multi-message CPA/CCA security definitions; (iii) extending ED to resist software forensics on a seized device; (iv) usability studies for key man-

agement and cover text selection; and (v) quantitative metrics for decoy message plausibility.

Ethical considerations. Second-order deniability raises dual-use concerns. PhantomCrypt is designed for journalists, activists, and whistleblowers in authoritarian settings, but could be exploited by malicious actors. We follow Albrecht et al. [3] in arguing that rigorous security analysis for vulnerable populations is both a technical and ethical imperative.

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A Correctness of PhantomCrypt

Proof (Proof that PhantomCrypt satisfies Definition 2). True message recovery. Given $\text{SK}_{\text{ie}} = (x_0, k)$, sk_{kem} , and T : (1) decapsulate C_{kem} to obtain K_{session} ; (2) decrypt C_1 to recover s_{new} ; (3) re-derive $k - 1$ shares from T using x_0 ; (4) reconstruct P through k points and evaluate $P(0) = m^*$. By correctness of AEAD and Lagrange interpolation, this succeeds with probability 1.

Decoy recovery. Given sk_{kem} and $\text{SK}_{\text{FBE},i}$: decapsulate, decrypt C_2 to recover c_{FBE} , apply FBE decryption. By correctness of FBE [1], m_i is recovered with probability 1. Note m^* is also recoverable via $\text{SK}_{\text{FBE},0}$, consistent with dual embedding.

Distinctness. Messages are elements of $\mathbb{F}_p$ chosen distinct at encryption time. $\square$

B PhantomCrypt Algorithms

We present the five core algorithms. For IE and FBE subroutines, see [1, 2].

Algorithm 1 PhantomCrypt.Setup($1^\lambda, L$)**Require:** Security parameter λ , cover text length L **Ensure:** PP, SK_{ie}, sk_{kem}

- 1: Select prime p with $\lceil \log_2 p \rceil = \lambda$
- 2: Initialize $\mathbb{F}_p$, H , PRNG with domain $[1, L]$
- 3: $(pk_{\text{kem}}, sk_{\text{kem}}) \leftarrow \text{KEM.KeyGen}(1^\lambda)$
- 4: $x_0 \xleftarrow{\$} \{0, 1\}^\lambda$ ▷ IE secret seed (pre-shared)
- 5: $k \xleftarrow{\$} \{3, \dots, \min(K_{\text{max}}, L/2)\}$ ▷ IE threshold (pre-shared)
- 6: $(c_{\text{FBE}}, r_{\text{FBE}}) \leftarrow \text{FBE.Init}(p, n_{\text{init}}, k_{\text{base}})$
- 7: PP $\leftarrow (p, H, \text{PRNG}, \text{KEM}, pk_{\text{kem}}, c_{\text{FBE}}, r_{\text{FBE}})$
- 8: SK_{ie} $\leftarrow (x_0, k)$
- 9: **return** (PP, SK_{ie}, sk_{kem})

Algorithm 2 PhantomCrypt.Enc(PP, SK_{ie}, sk_{kem} , m^*, T)**Require:** PP, SK_{ie} = (x_0, k) , sk_{kem} , $m^* \in \mathbb{F}_p$, T **Ensure:** Ciphertext C , FBE key for m^*

- 1: $(s_{\text{new}}, x_{\text{new}}, S) \leftarrow \text{IE.Enc}(PP, x_0, k, m^*, T)$ ▷ IE layer
- 2: $(c_{\text{FBE}}, SK_{\text{FBE},0}) \leftarrow \text{FBE.Enc}(PP, c_{\text{FBE}}, m^*, r_{\text{FBE}}, k)$ ▷ FBE layer
- 3: $(C_{\text{kem}}, K_{\text{session}}) \leftarrow \text{KEM.Encaps}(pk_{\text{kem}})$ ▷ Hybrid envelope
- 4: $IV_1 \xleftarrow{\$} \{0, 1\}^{96}$; $IV_2 \xleftarrow{\$} \{0, 1\}^{96}$
- 5: $C_1 \leftarrow \text{AE.Enc}(K_{\text{session}}, s_{\text{new}}, IV_1)$
- 6: $C_2 \leftarrow \text{AE.Enc}(K_{\text{session}}, \text{Serialize}(c_{\text{FBE}}), IV_2)$
- 7: **return** $((C_{\text{kem}}, C_1, C_2, IV_1, IV_2), SK_{\text{FBE},0})$

Algorithm 3
PhantomCrypt.AddDecoy**Require:** PP, sk_{kem} , C , m_{decoy} **Ensure:** Updated C' , new FBE key

- 1: $K_{\text{session}} \leftarrow \text{KEM.Decaps}(sk_{\text{kem}}, C_{\text{kem}})$
- 2: $c_{\text{FBE}} \leftarrow \text{Deserialize}(\text{AE.Dec}(K_{\text{session}}, C_2, IV_2))$
- 3: **if** decryption fails **then return** $\perp$
- 4: **end if**
- 5: $(c'_{\text{FBE}}, SK_{\text{FBE},\text{new}}) \leftarrow \text{FBE.Enc}(PP, c_{\text{FBE}}, m_{\text{decoy}}, r_{\text{FBE}}, k)$
- 6: $IV'_2 \xleftarrow{\$} \{0, 1\}^{96}$
- 7: $C'_2 \leftarrow \text{AE.Enc}(K_{\text{session}}, \text{Serialize}(c'_{\text{FBE}}), IV'_2)$
- 8: **return** $((C_{\text{kem}}, C_1, C'_2, IV_1, IV'_2), SK_{\text{FBE},\text{new}})$

Algorithm 4
PhantomCrypt.DecDecoy**Require:** PP, SK_{FBE, i} = (I_c, I_r, pos) , C , sk_{kem} **Ensure:** m_i or $\perp$

- 1: $K_{\text{session}} \leftarrow \text{KEM.Decaps}(sk_{\text{kem}}, C_{\text{kem}})$
- 2: $c_{\text{FBE}} \leftarrow \text{Deserialize}(\text{AE.Dec}(K_{\text{session}}, C_2, IV_2))$
- 3: **if** decryption fails **then return** $\perp$
- 4: **end if**
- 5: $m_i \leftarrow \text{FBE.Dec}(c_{\text{FBE}}, SK_{\text{FBE},i}, r_{\text{FBE}})$
- 6: **return** m_i

Algorithm 5 PhantomCrypt.DecTrue(PP, SK_{ie}, sk_{kem}, C, T)

Require: PP, SK_{ie} = (x₀, k), sk_{kem}, C, T

Ensure: m* or ⊥

- 1: K_{session} ← KEM.Decaps(sk_{kem}, C_{kem})
- 2: s_{new} ← AE.Dec(K_{session}, C₁, IV₁)
- 3: **if** decryption fails **then return** ⊥
- 4: **end if**
- 5: {(x_i, s_i)}_{i=1}^{k-1} ← IE.DeriveShares(x₀, k, T, H, PRNG)
- 6: x_{new} ← IE.GetAbscissa(x₀, k, H)
- 7: m* ← LagrangeInterpolate({(x_i, s_i)}_{i=1}^{k-1} ∪ {(x_{new}, s_{new})})
- 8: **return** m*

▷ m* = P(0)
